Data S5. List of morphological characters.

This list is edited follows the scoring methods of Seiffert et al. (2020) (8). Sources of characters: Ross et al. (1998) (26); Yoder (1994) (105); Boyer et al. 2016 (106); Gebo et al. 2001 (107); Seiffert and Simons 2001 (108); Yapuncich et al. 2017 (109); Anemone and Covert, 2000 (109); Fleagle and Kay, 1983 (111); Maier, 2008 (111); Cartmill and Kay , 1978 (113); Masters and Brothers, 2002 (114); Luckett, 1976 (115). When no reference for the character is given, the character is from Seiffert et al. 2020. References may be abbreviated: For example, RCr refers to Ross et al 1998 Cranial characters; Y refers to Yoder, 1994, etc.

1. lower incisor number (modified from Ross et al. 1998 character i1)
    0 three incisors in each quadrant
    1 two incisors in each quadrant
    2 one incisor in each quadrant
    3 no incisors present

2. lower incisor occlusal arrangement (Ross et al. 1998 character i2)
    0 arcuate battery in occlusal view ("U"-shaped arcade)
    1 incisor cusp tips staggered ("V"-shaped arcade)

3. toothcomb (modified from Ross et al. 1998 character i13)
    0 lower canine is not incorporated into a toothcomb
    1 lower canine is incorporated into a toothcomb

4. lower i1-2 area relative to lower m1 area (ratio of i1 + i2 area / m1 area) (Ross et al. 1998 character i5)
    0 small (=<0.69)
    1 intermediate (=>0.70, =<1.07)
    2 large (>1.07)

5. i1:i2 proportions (Ross et al. 1998 character i6)
    0 i1:i2 area ratio <0.65
    1 i1:i2 area ratio =>0.65, <0.82
    2 i1:i2 area ratio =>0.83, <1.00
    3 i1:i2 area ratio =>1.01,<1.25
    4 i1:i2 area ratio =>1.25

6. i2 cross-sectional shape (ratio of mesiodistal length to buccolingual breadth) (Ross et al. 1998 character i8)
    0 ratio =>0.64
    1 ratio =<0.64

7. lower incisor crown height (Ross et al. 1998 character i9)
    0 low-crowned
    1 0/2
    2 moderately high-crowned
    3 2/4
    4 high-crowned

8. lower incisor root orientation (Ross et al. 1998 character i11)
    0 erect or vertical
    1 0/2
    2 slightly procumbent
    3 2/4
    4 very procumbent

9. incisor crown orientation (Ross et al. 1998 character i12)
    0 erect or vertical
    1 0/2
    2 procumbent

3 2/4
4 very procumbent

10. first lower incisor crown shape (Ross et al. 1998 character i14)
0 spatulate
1 pointed or lanceolate

11. i2 heel development (Ross et al. 1998, character i15)
0 heel absent
1 heel present

12. lower first incisor lingual cingulum (Ross et al. 1998 character i17)
0 absent to weak
1 0/2
2 strong but incomplete
3 2/4
4 strong and complete

13. i1 to m1 area (Ross et al. 1998 character i19)
0 i1 area << m1 area
1 0/2
2 i1 area < or = m1 area
3 2/4
4 i1 > m1

14. female canine maximum cross-sectional area relative to first lower molar area (Ross et al. 1998 character c1, modified)
0 c1 area/m1 area < 0.40
1 0/2
2 c1 area/m1 area =>0.4, <0.8
3 2/4
4 c1 area/m1 area =>0.8, <=1.2

15. canine dimorphism (square root of male lower canine area / square root of female lower canine area) (Ross et al. 1998 character c2)
0 <1.07
1 >=1.07, <1.17
2 >=1.17

16. lower canine cross-sectional shape (Ross et al. 1998 character c3)
0 rounded oval
1 mesiodistally compressed
2 buccolingually compressed

17. orientation of lower canine root
0 oriented in line with long axis of crown
1 mesially inflected

18. lower canine lingual crest development (Ross et al. 1998 character c4)
0 rounded
1 0/2
2 sharp

19. lower canine paracristid (Ross et al. 1998 character c5, modified)
0 oblique to occlusal plane
1 nearly horizontal to occlusal plane

2 forms part of cropping mechanism with i1-2
3 oriented in line with buccal face of adjacent incisor

20. lower canine height (Ross et al. 1998 character c6, modified)
0 low, squat
1 0/2
2 narrow, short
3 tall, at or above toothrow
4 high crowned but very procumbent

21. p1 presence (Ross et al. 1998 character p1, modified)
0 present
1 0/2
2 absent

22. p2 roots/presence (Ross et al. 1998 character p3, modified)
0 two
1 0/2
2 one
3 2/4
4 p2 absent

23. p2 metaconid (Ross et al. 1998 character p15)
0 metaconid absent or indistinct
1 0/2
2 metaconid is small
3 2/4
4 metaconid is large

24. p2 protoconid height and shape (Ross et al. character p13)
0 extremely short, shorter than p3
1 not projecting, in line with p3
2 slender, projects above protoconids of p3-4
3 massive, projects above protoconids of p3-4

25. p2 buccal cingulum development (Ross et al. character p28)
0 absent
1 0/2
2 incomplete, broken at protoconid and hypoconid
3 2/4
4 complete

26. p3 paraconid (Ross et al. 1998 character p6, modified)
0 present, large
1 0/2
2 present, small
3 2/4
4 negligible or absent

27. p3 lingual cingulum
0 present
1 0/2
2 absent

28. p3 metaconid size (Ross et al. 1998 character p16)
0 absent

1 0/2
2 present, small
3 2/4
4 present, large

29. p3 lateral protocristid orientation (Ross et al. 1998, character p22, modified)
0 transversely oriented
1 0/2
2 distally or distolingually oriented
3 2/4
4 absent

30. p3 roots (Ross et al. 1998 character p4, in part, modified)
0 one
1 0/2
2 two

31. p3 root orientation (contains information like Ross et al. 1998 character p45)
0 mesial root placed directly mesial to distal root
1 0/2
2 mesial root placed lateral to distal root

32. p3 hypoconid (contains information like that of Ross et al. 1998 character p25)
0 large and well-defined, projects above other talonid cusps
1 0/2
2 very small, absent, or incorporated into talonid as an indistinct cusp

33. p3 entoconid and lingual talonid crest (Ross et al. 1998 character p19)
0 absent
1 0/2
2 lingual talonid crest present but an entoconid does not stand out above it
3 2/4
4 entoconid is a small discrete cusp

34. p3-4 protoconid height (Ross et al. 1998 character p41)
0 p3 much lower than p4
1 p3 slightly lower than p4
2 p3 equal in height to p4
3 p3 higher than p4

35. p3 area/p4 area (Ross et al. 1998 character p42)
0 <0.45
1 0.45-0.59
2 0.60-0.69
3 0.70-0.79
4 >0.80

36. p4 root number (Ross et al. 1998 character p4, in part)
0 one
1 two

37. p4 metaconid size (Ross et al. 1998 character p17)
0 absent or indistinct
1 0/2
2 small
3 2/4

4 large

38. spacing of p4 metaconid and protoconid (Ross et al. 1998 character p14)
    0 close to protoconid
    1 0/2
    2 widely spaced from protoconid

39. mesiodistal position of p4 metaconid with respect to protoconid
    0 metaconid is distal to protoconid
    1 0/2
    2 metaconid is placed transverse to protoconid

40. p4 paraconid size (Ross et al. 1998 character p7, modified)
    0 p4 paraconid is present and large
    1 0/2
    2 p4 paraconid is present and small
    3 2/4
    4 p4 paraconid is very small or absent

41. p4 paraconid position (Ross et al. 1998 character p9)
    0 mesial to protoconid
    1 mesiolingual, between protoconid and metaconid
    2 mesial to metaconid; widely spaced from metaconid

42. p4 lateral protocristid morphology (contains information like Rp21-42 and Rp23-44)
    0 protocristid poorly developed or absent
    1 0/2
    2 distolingually oriented protocristid present
    3 2/4
    4 trenchant and tranversely oriented protocristid present

43. p4 premetacristid (Ross et al. 1998 character p18, modified)
    0 premetacristid absent or poorly developed
    1 0/2
    2 premetacristid present

44. p4 postprotocristid (Ross et al. 1998 character p36)
    0 weak or absent
    1 0/2
    2 moderate
    3 2/4
    4 very strong

45. p4 postmetacristid (contains information like Ross et al. 1998 character p37)
    0 weak or absent
    1 0/2
    2 moderate
    3 2/4
    4 very strong

46. p4 entoconid and lingual talonid crest (Ross et al. 1998 character p20)
    0 absent or a trace
    1 0/2
    2 lingual talonid crest present but an entoconid does not stand out above it
    3 2/4
    4 entoconid is a small discrete cusp

47. height of p4 talonid
    0 no aspect of the talonid is at level of m1 trigonid basin
    1 0/2
    2 hypoconid extends to level of m1 trigonid basin
    3 2/4
    4 entire talonid basin extends to level of m1 trigonid basin

48. p4 cristid obliqua development (Ross et al. 1998 character p11)
    0 absent
    1 0/2
    2 weak
    3 2/4
    4 strong

49. position of p4 hypoconid or distal terminus of p4 cristid obliqua (Ross et al. 1998 character p26, modified)
    0 buccal to protoconid
    1 distal to protoconid
    2 between protoconid and metaconid
    3 distal to metaconid

50. p4 hypocristid shearing development (Ross et al. 1998 character p27)
    0 absent
    1 0/2
    2 weak
    3 2/4
    4 strong

51. lower premolar crowding (Ross et al. 1998 character p5)
    0 no crowding
    1 0/2
    2 slightly crowded
    3 2/4
    4 very crowded

52. lower premolar inflation (Ross et al. 1998 character p29)
    0 not basally inflated
    1 0/2
    2 slightly basally inflated
    3 2/4
    4 very basally inflated

53. premolar orientation (Ross et al. 1998 character p33)
    0 crown bases vertical in lateral perspective
    1 0/2
    2 slightly oblique
    3 2/4
    4 strongly oblique, projecting mesially

54. p4 exodaenodonty (Ross et al. 1998 character p30)
    0 not exodaenodont
    1 slightly exodaenodont
    2 very exodaenodont

55. buccal cingulum on p4 (Ross et al. 1998 character p34)
    0 absent or indistinct

1 0/2
2 present and distinct

56. p4 lingual cingulum
0 absent or discontinuous
1 0/2
2 present

57. p4 md length/bl width (Ross et al. 1998 character p43)
0 <0.95
1 >0.96, <1.14
2 >1.15, <1.35
3 >1.36, <1.46
4 >1.47

58. p4 area relative to m1 area (Ross et al. 1998 character p44)
0 <0.62
1 >0.63, <0.72
2 >0.73, <0.82
3 >0.83, <0.92
4 >0.93

59. m1 length/m1 width (Ross et al. 1998 character m55)
0 1.0-1.15
1 1.16-1.22
2 1.23-1.32
3 >1.33

60. m1 area (Ross et al. 1998 character m54)
0 1.10-3.00 mm
1 3.10-5.00 mm
2 5.10-7.00 mm
3 7.10-9.00 mm
4 >9.10 mm

61. m1 paraconid size (contains information like Ross et al. 1998 character m20)
0 absent or crestiform
1 0/2
2 present, small
3 2/4
4 present, large

62. m1 paraconid position (Ross et al. 1998 character m8)
0 mesial to protoconid
1 mesiolingual, equidistant between protoconid and metaconid
2 1/3
3 mesial to metaconid but widely spaced from it
4 twinned with metaconid

63. m1 parastylid (Ross et al. 1998 character m11)
0 absent
1 0/2
2 present

64. m1 premetacristid (contains information like Ross et al. 1998 character m18)
0 premetacristid absent or indistinct

1 0/2
2 premetacristid present

65. m1 metaconid position (Ross et al. 1998 character m19)
0 approximately transverse to protoconid
1 0/2
2 distal to protoconid

66. m1 trigonid height (ratio of trigonid height to talonid height) (Ross et al. 1998 character m16)
0 higher than talonid (=>1.20)
1 0/2
2 slightly higher than talonid (=>1.10, <1.20)
3 2/4
4 trigonid and talonid of similar height

67. m1 cristid obliqua orientation (Ross et al. 1998 character m32)
0 reaches trigonid wall at a point distal to protoconid
1 0/2
2 reaches trigonid wall at a point distolingual to protoconid
3 2/4
4 reaches trigonid wall at a point distal to metaconid

68. m1 cristid obliqua terminus (Ross et al. 1998 character m34)
0 runs to base of trigonid
1 0/2
2 runs part way up the distal trigonid wall
3 2/4
4 connects with protoconid tip or protocristid
5 4/6
6 connects with metaconid

69. hypoconulid position on m1 (contains information like Ross et al. 1998 character m30)
0 twinned with, or most closely situated next to, entoconid
1 0/2
2 central
3 2/4
4 closer to hypoconid than to entoconid

70. m1 hypoconulid size (Ross et al. 1998 character m27)
0 large
1 moderate
2 small
3 absent

71. m1 cristid obliqua development (Ross et al. 1998 character m31, in part)
0 weak (rounded)
1 0/2
2 strong (trenchant)

72. m1-2 hypocristid development (Ross et al. 1998 character m38)
0 absent or indistinct
1 0/2
2 weak
3 2/4
4 strong

73. m1-2 buccal cingulum development (Ross et al. 1998 character m45)
    0 absent to trace
    1 0/2
    2 partial, discontinuous
    3 2/4
    4 complete

74. m1-2 talonid -- lingual configuration (Ross et al. 1998 character m40)
    0 open
    1 0/2
    2 notched lingually or closed

75. m2 paraconid size (contains information like Ross et al. 1998 character m20)
    0 absent
    1 0/2
    2 present, small
    3 2/4
    4 present, large

76. m2 paraconid position (Ross et al. 1998 character m9)
    0 mesial to protoconid
    1 0/2
    2 mesiolingual, approximately equidistant between protoconid and metaconid
    3 2/4
    4 mesial to metaconid but widely spaced from it
    5 twinned with metaconid

77. m2 trigonid width (ratio of buccolingual breadths of trigonid and talonid) (Ross et al. 1998 character m6)
    0 much wider than talonid (=>1.11)
    1 0/2
    2 width similar (<1.11,>0.90)
    3 2/4
    4 much narrower than talonid (=<0.90)

78. m2 cristid obliqua orientation (Ross et al. 1998 character m33)
    0 reaches trigonid wall at a point distal to protoconid
    1 0/2
    2 reaches trigonid wall at a point distolingual to protoconid
    3 2/4
    4 reaches trigonid wall at a point distal to metaconid

79. m2 cristid obliqua terminus (Ross et al. 1998 character m35)
    0 runs to base of trigonid
    1 0/2
    2 runs part way up distal trigonid wall
    3 connects to protoconid tip or protocristid
    4 3/5
    5 connects with metaconid

80. m2 cristid obliqua development (Ross et al. 1998 character m31, in part)
    0 weak (rounded)
    1 0/2
    2 strong (trenchant)

81. position of entoconid relative to hypoconid on m2 (contains information like Ross et al. 1998 character m57)
    0 mesial to hypoconid

1 0/2
2 transverse to hypoconid
3 2/4
4 distal to hypoconid

82. hypoconulid position on m2 (contains information like Ross et al. 1998 character m30)
0 twinned with, or most closely situated next to, entoconid
1 0/2
2 central
3 2/4
4 closer to hypoconid than to entoconid

83. m2 hypoconulid size (Ross et al. 1998 character m28)
0 large
1 moderate
2 small
3 absent

84. morphology of distolingual aspect of m2 (contains information like Ross et al. 1998 characters m26 and m41)
0 no postentoconid sulcus
1 hypoconulid closely situated next to entoconid and postentoconid sulcus present, entoconid confluent with distal talonid wall
2 distolingual fovea present, entoconid confluent with posterior talonid wall
3 distolingual fovea present, entoconid mesiodistally abbreviated and not confluent with posterior talonid wall

85. accessory cusps on distolingual aspect of m2
0 absent
1 variably present

86. cusp basal inflation on m1-2 (contains information like Ross et al. 1998 character m17 and m44)
0 crestiform
1 0/2
2 moderate basal inflation
3 2/4
4 cusp bases bulbous

87. m2 length/m3 length (Ross et al. 1998 character m53)
0 m3 much longer than m2 (0.71-0.80)
1 m3 longer than m2 (0.81-0.90)
2 m3 slightly longer or equal in length to m2 (0.91-1.00)
3 m3 slightly shorter than m2 (1.01-1.12)
4 m3 much shorter than m2 (=>1.13) or absent

88. m3 paraconid size
0 absent
1 0/2
2 present, small
3 2/4
4 present, large

89. m3 paraconid position (Ross et al. 1998 character m10)
0 mesial to protoconid
1 0/2
2 mesiolingual, between protoconid and metaconid
3 2/4

4 mesial to metaconid but widely spaced from it
5 4/6
6 twinned with metaconid

90. m3 cristid obliqua terminus (Ross et al. 1998 character m36, modified)
0 absent or indistinct
1 runs to base of trigonid
2 runs part way up distal trigonid wall
3 connects with protoconid tip or protocristid
4 connects with metaconid

91. m3 trigonid width (based on relative buccolingual breadths) (Ross et al. 1998 character m7)
0 =>1.20
1 0/2
2 <1.20->1.05
3 2/4
4 <1.05

92. m3 entoconid
0 present and distinct
1 0/2
2 greatly reduced or absent

93. morphology of m3 hypoconulid
0 lobate
1 0/2
2 not lobate

94. position of hypoconulid on m3
0 centrally located
1 twinned with, or situated closest to, entoconid
2 buccally located, closer to hypoconid

95. m3 hypoconulid (Ross et al. 1998 character m13)
0 absent
1 0/2
2 single
3 2/4
4 double, or additional cusp distal to entoconid that is independent of distal lobe

96. m3 root number (Ross et al. 1998 character m4)
0 one
1 two

97. molar centroconids (contains information like Ross et al. 1998 character m37)
0 absent
1 present

98. sulcus between metaconid and protoconid on m1-2 (contains information like Ross et al. 1998 character m22 and m23)
0 absent
1 0/2
2 present; protocristids poorly developed or absent

99. lower molar metastylids (Ross et al. 1998 character m12)
0 absent

1 0/2
2 trenchant postmetacristid or incipient metastylid
3 2/4
4 present, large

100. molar enamel crenulation (Ross et al. 1998 character m15)
0 smooth
1 0/2
2 slightly crenulated
3 2/4
4 highly crenulated

101. interincisal diastema (RCr42-224)
0 very broad
1 narrow

102. I1-I2 interstitial contact (RI1-114)
0 absent, teeth widely spaced
1 0/2
2 present as a narrow contact
3 2/4
4 I2 tightly packed against I1, I1 paracrista abbreviated

103. I1:I2 area (RI4-117)
0 areas approximately equal or I1 smaller than I2 (=<1.00)
1 0/2
2 I1 slightly larger than I2 (>1.00,<1.40)
3 2/4
4 I1 much larger than I2 (>1.40)

104. I1-2 lingual cingulum (RI11-124)
0 weak, discontinuous
1 0/2
2 moderate, continuous
3 2/4
4 strong

105. I1-I2 buccal cingulum (RI13-126)
0 absent
1 present

106. I1 occlusal shape (mesiodistal length/buccolingual breadth) (RI6-119)
0 rounded oval (<1.05)
1 0/2
2 buccolingually compressed (>1.05, <1.30)
3 2/4
4 extremely compressed (>1.30)

107. I1 crown shape (RI8-121)
0 spatulate (no apparent occlusal cusp, mesial and distal edges continuous and rounded)
1 0/2
2 semi-spatulate (central cusp present, but blunt with discernable mesial and distal occlusal crests)
3 2/4
4 central occlusal cusp pointed, occlusal edges steep

108. I1 size (I1 area:M1 area) (RI5-118)

0 small (=<0.50)
1 moderate (>0.5, <0.56)
2 large (>0.56)

109. I1 occlusal edge orientation (spatulate incisors only) (RI10-123)
0 occlusal edge approximately orthogonal to long axis of root
1 occlusal edge wears at a steep angle to long axis of root
2 crown with pronounced mesial asymmetry (=mesial process) in unworn state

110. I1 basal lingual cusp (RI12-125)
0 absent
1 present

111. I2 occlusal shape (mesiodistal length/buccolingual breadth) (RI7-120)
0 rounded oval (=<1.05)
1 0/2
2 slightly buccolingually compressed (>1.05, <1.30)
3 2/4
4 extremely buccolingually compressed (=>1.30)

112. I2-C diastema (RI3-116)
0 present
1 0/2
2 absent

113. upper incisor number (Y43)
0 three in each quadrant
1 two in each quadrant
2 one or none in each quadrant

114. upper canine shape (contains information like RC1-127)
0 round in cross-section
1 0/2
2 oval
3 2/4
4 strongly compressed buccolingually

115. rotation of upper canine
0 long axis of canine oriented in line with adjacent premolar
1 canine is slightly rotated internally
2 canine is strongly rotated internally

116. C1 mesial groove (females) (RC3-129)
0 shallow or absent
1 deep

117. C1 lingual cingulum (RC4-130)
0 weak or absent
1 0/2
2 strong
3 2/4
4 very strong

118. upper canine occlusion (RC2-128)
0 wears against p1-2
1 wears against p2

2 wears against p2-3
3 wears against p3

119. P1 presence (contains information from Y45)
0 present
1 absent

120. P2 root number/presence (RP1-131, modified)
0 three
1 two
2 one
3 P2 absent

121. P2 occlusal outline (RP6-136)
0 triangular
1 suboval with the long axis buccolingual
2 suboval with the long axis mesiodistal
3 round

122. P2 protocone (RP12-142, modified)
0 present
1 absent

123. height of P2 relative to P3*
0 smaller than, or subequal in height (P2 height/P3 height <=1.1)
1 0/2
2 P2 slightly taller than P3 (P2 height/P3 height =>1.11, <1.5)
3 2/4
4 P2 much taller than P3 (P2 height/P3 height >1.51)

124. P2 area relative to P3 area (contains information like that in RP4-134)
0 much smaller than P3 (<=0.85) (if P2 is absent, taxon is assigned state 0)
1 approximately the same size, or only slightly smaller, than P3 (>0.85, <0.95)
2 much larger than P3 (>=.95)

125. P3 distal crown margin (RP18-148)
0 smoothly rounded
1 0/2
2 waisted between buccal and lingual cusps

126. P3 protocone (RP9-139, modified)
0 absent
1 0/2
2 present, small
3 2/4
4 present, distinct

127. P3 hypocone (RP13-143, in part)
0 absent
1 0/2
2 present

128. P3 metacone (RP20-150)
0 absent
1 0/2
2 present

129. P3 root number (RP2-132)
    0 one
    1 0/2
    2 two
    3 2/4
    4 three

130. P4 parastyle (RP15-145, modified)
    0 present
    1 0/2
    2 absent

131. P4 metastyle (RP16-146, modified)
    0 absent
    1 0/2
    2 present

132. P3-4 trigon/talon proportions (RP8-138)
    0 trigon>=talon
    1 trigon<talon

133. P3-4 buccal cingulum development (RP21-151)
    0 absent or weak
    1 0/2
    2 strong

134. upper premolar paraconules (contains information like RP14-144)
    0 absent or indistinct
    1 transverse crest connecting protocone to paracone
    2 present

135. P4 lingual cingulum (contains information like RP19-149)
    0 absent
    1 0/2
    2 present

136. P4 metacone (RP10-140)
    0 absent
    1 0/2
    2 present

137. P4 protocone (RP11-141)
    0 low relative to paracone
    1 high relative to paracone

138. P4 hypocone (RP13-143, in part)
    0 absent
    1 0/2
    2 present

139. P4 pericone
    0 absent
    1 present

140. P4 metaconules

0 absent
1 0/2
2 present

141. P4 postprotocrista (RP17-147, in part)
0 absent
1 weak, short
2 strong
3 strong; secondary postprotocrista directed toward metacone

142. P4 occlusal outline (RP7-137)
0 triangular
1 suboval
2 1/3
3 squared

143. P4 root number (RP3-133)
0 one
1 two
2 three

144. P4 area:M1 area (RP5-135)
0 P4<<M1 (<=0.66)
1 P4<M1 (>0.66,<=0.76)
2 P4=M1 (0.77-1.05)
3 P4>M1 (>1.06)

145. M1 pericone (RM20-170, modified)
0 absent
1 0/2
2 small
3 2/4
4 large

146. M1 area:M2 area (RM4-155)
0 M1>M2 (<1.40,>1.0)
1 0/2
2 M1=<M2 (=<1.0)

147. M1 mesiobuccal fovea delimited by trenchant parahypocrista
0 absent
1 0/2
2 present

148. M1-2 paraconule size (RM37-180)
0 absent
1 0/2
2 small-moderate (smaller than paracone)
3 2/4
4 large (nearly as large as protocone)

149. M1-2 paraconule position (RM15-165)
0 attached to preprotocrista
1 0/2
2 not attached to preprotocrista

150. M1 hypocone size (RM10-160)
0 large
1 0/2
2 small
3 2/4
4 absent

151. M1-2 hypocone position (RM12-162)
0 distal, slightly lingual to protocone
1 0/2
2 distal, far lingual to protocone

152. convex distal lobe for M1-2 hypocone
0 absent
1 0/2
2 present, distal crown margin weakly concave
3 2/4
4 present, distal crown margin deeply notched

153. lingual extension of M1 hypocone lobe
0 lobe does not extend lingually far beyond protocone
1 0/2
2 lobe does extend far lingually past protocone

154. M2 prehypocrista development (RM13-163)
0 absent
1 0/2
2 weak
3 2/4
4 strong -- reaches to postprotocrista, encloses the talon lingually

155. M1-2 *Nannopithex* fold (RM5-156)
0 absent
1 0/2
2 weak
3 2/4
4 strong

156. M1-2 postprotocrista development (RM18-168)
0 strong, runs to base of metaconule or metacone
1 0/2
2 strong but short, does not reach to base of metacone
3 2/4
4 absent

157. M1-2 metaconule size (RM16-166)
0 absent
1 small
2 moderate
3 large

158. M1-2 premetaconule cristae (RM27-173)
0 absent or weak
1 0/2
2 strong

159. M1-2 postmetaconule cristae (RM28-174)*
0 absent or weak
1 0/2
2 strong

160. M1-2 lateral posterior transverse crista development (RM19-169)
0 sharp
1 0/2
2 indistinct

161. shape of M2 centrocrista
0 straight, aligned with mesiodistal plane
1 0/2
2 crests are more buccally oriented, meet at an angle
3 2/4
4 crests meet at a sharp angle or form a mesostyle

162. buccal cingulum on M1-2 (contains information like RM24-172)
0 absent
1 present but poorly developed
2 present and distinct
3 extensive buccal "shelf"

163. M1-2 root number (RM1-152)
0 three or more, three or more
1 three, two

164. M2 shape (bl/md) (RM3-154)
0 very broad (>1.65)
1 0/2
2 broad (<1.65,>1.30)
3 2/4
4 squared (=<1.30)

165. M2 pericone
0 absent
1 0/2
2 present

166. M2 hypocone size (RM11-161)
0 large
1 0/2
2 small
3 2/4
4 absent

167. M2 postmetacrista
0 short, indistinct, or absent
1 long, trenchant, and labially extended

168. M2 buccal expansion of paracone (RM33-177)
0 no expansion
1 0/2
2 slight expansion
3 2/4
4 considerable expansion

169. M1-3 lingual cingulum development (RM22-171)
    0 absent
    1 0/2
    2 weak, broken
    3 2/4
    4 strong, complete

170. M1-3 anterior cingulum (RM44-181)
    0 strong, complete, long
    1 0/2
    2 strong, short
    3 2/4
    4 weak or absent

171. molar protocone lingual inflation (RM31-176)
    0 not inflated
    1 0/2
    2 slightly inflated
    3 2/4
    4 very inflated

172. M3 metacone (RM34-178)
    0 absent or very small
    1 0/2
    2 moderate (but smaller than paracone)
    3 2/4
    4 large (equal to paracone)

173. M3 paraconule (RM30-175)
    0 absent
    1 0/2
    2 small-moderate
    3 2/4
    4 large

174. M3 hypocone (RM36-179)
    0 absent or very small
    1 0/2
    2 small
    3 2/4
    4 large

175. M3 size relative to M1 (RM46-182)
    0 very small (half the size of M1 or less)
    1 0/2
    2 small (two-thirds)
    3 2/4
    4 large (approximately as large)

176. M3 root number (RM2-153)
    0 three
    1 0/2
    2 two
    3 2/4
    4 one

177. mandibular symphyseal fusion (Y57,RCr39-221)
0 absent
1 0/2
2 present

178. shape of the mandibular angle (Y40)
0 sharply hooked
1 0/2
2 smoothly rounded
3 2/4
4 rounded but expanded posteriorly

179. shape of masseteric fossa
0 shallow
1 0/2
2 deeply excavated

180. mandibular depth (Y39,RCr45-227)
0 shallow (less than 1.8 times as deep @ m2 as md length of m2)
1 0/2
2 deep (>1.8 times as deep @ m2 as md length of m2, <2.19)
3 2/4
4 very deep (more than 2.2 times as deep @ m2 as md length of m2)

181. height of coronoid process relative to condyle (RCr43-225)
0 very high above
1 slightly above or equal

182. condyle height relative to toothrow (RCr44-226)
0 at level of toothrow
1 slightly above
2 well above toothrow

183. scapular morphology (mediolateral width/inferosuperior height) (contains information like that of Y64)
0 <1.25
1 <1.5, >1.25
2 >1.5, <1.75
3 >1.75

184. length of scapular coracoid process (coracoid process length/glenoid fossa height)
0 <=.90
1 >.9, <=1.0
2 >1.0, <=1.1
3 >1.1

185. relative sizes of infraspinous and supraspinous fossae (Y66)
0 supraspinous fossa larger than infraspinous fossa
1 fossae roughly equivalent
2 infraspinous fossa approximately two times as large as the supraspinous fossa
3 infraspinous fossa approximately three times as large as the supraspinous fossa

186. position of humeral head relative to tubercles
0 humeral head is expanded dorsal to tubercles
1 0/2
2 humeral head and tubercles are approximately equal in height

3 2/4
4 tubercles are expanded above humeral head

187. morphology of deltopectoral crest (RH16-247)
0 prominent
1 0/2
2 low

188. shape of intertubercular sulcus (contains information like RH15-246)
0 narrow and deep
1 0/2
2 narrow and shallow
3 2/4
4 shallow and wide

189. teres major insertion
0 absent or poorly defined
1 0/2
2 elongate well-defined crest
3 2/4
4 well-developed tubercle

190. supinator crest (Y76, contains information like RH13-244)
0 prominent
1 low

191. brachialis flange (RH14-245)
0 broad
1 0/2
2 moderate
3 2/4
4 narrow

192. presence/mediolateral position of entepicondylar foramen (Seiffert et al., 2000)
0 foramen is placed far proximal to trochlea
1 lateral wall of medial strut defining foramen is confluent with medial edge of the trochlea
2 0/2
3 foramen more medial, and lateral wall of medial strut defining foramen is not confluent with medial edge of the trochlea
4 absent

193. dorsal placement of medial epicondyle (RH9-240)
0 parallel
1 0/2
2 slight dorsal angle
3 2/4
4 large dorsal angle

194. medial epicondyle size (RH8-239)
0 reduced
1 0/2
2 prominent

195. dorsoepitrochlear fossa (RH11-242)
0 present
1 0/2

2 small, shallow
3 2/4
4 absent

196. shape of humeral trochlea (contains information like RH1-232)
0 cylindrical, distomedial aspect of trochlea is straight
1 0/2
2 conical, medial aspect of trochlea flares distally

197. relative heights of medial and lateral edges of the humeral trochlea (RH2-233)
0 subequal
1 0/2
2 medial edge more flared than lateral edge

198. shape of dorsal aspect of distal humeral articulation (contains information like that of RH10-241)
0 no pronounced lips on dorsal trochlear edges
1 0/2
2 both medial and lateral edges pronounced
3 very pronounced lateral lip

199. trochlear-capitular junction
0 trochlea and capitulum are confluent, with no distinct trochleo-capitular ridge
1 0/2
2 lateral aspect of trochlea is offset from capitulum by a weak ridge
3 2/4
4 lateral aspect of trochlea is separated from capitulum by a deep gutter

200. capitular shape
0 globular
1 0/2
2 ovoid

201. capitular tail
0 elongate and distinct
1 short or absent
2 1/3
3 proximodistally tall capitular flange is present

202. relative width of capitulum (contains information like RH5-236)
0 ventral articular width/capitular width >2.5
1 0/2
2 <2.5, >2.0
3 2/4
4 <2.0

203. olecranon fossa morphology (Y78, contains information like RH12-243)
0 deep, open and unossified
1 0/2
2 moderately deep, thinly ossified and nearly transparent
3 2/4
4 shallow, robustly ossified

204. humerofemoral index
0 <59
1 60-69
2 70-79

3 80-89
4 90-100

205. size of centrale, orientation of centrale trapezoid facet, and articulation with hamate (modified from RW1-249)
0 facet faces distally, no articulation with hamate
1 facet faces distoradially, articulation with hamate
2 centrale not present as an independent element

206. ulnar-pisiform articulation (modified from RW2-250)
0 no ulnar-pisiform articulation
1 facet on pisiform for ulnar styloid process is smaller than, or roughly equal in size to, that for triquetrum
2 facet on pisiform for ulnar styloid process is much enlarged and deeply excavated

207. styloid process (Y80)
0 no discernable styloid process
1 present but only moderately developed
2 long and well-developed

208. second digit of hand (Y81)
0 of normal length
1 drastically reduced

209. pollical divergence angle (Boyer et al., 2016)
0 0-15 degrees
1 0/2
2 16-30 degrees
3 2/4
4 >31 degrees

210. nail/claw on second pedal digit (Y82,RO2-283)
0 nail present
1 claw present

211. prehallux (RO3-284)
0 present
1 absent

212. astragalar width/astragalar length (Gebo et al., 2001, character 6)
0 <50
1 >50, <60
2 >60, <70
3 >70

213. astragalar body height (lateral body height/mid-trochlear width; Gebo et al., 2001, character 4)
0 <100
1 0/2
2 100-120
3 2/4
4 >120

214. elevated lateral trochlear margin of astragalus (Seiffert & Simons, 2001)
0 absent
1 0/2
2 present

215. astragalar cotylar fossa (Seiffert & Simons, 2001)

0 shallow
1 0/2
2 moderate
3 2/4
4 deep, medially projecting

216. medial tibial facet/ectal facet area ratio (Boyer et al., 2016)
0 < -0.151
1 between -0.15 and 0
2 between 0 and 0.15
3 > 0.16

217. medial tibial facet perimeter/medial tibial facet area (Boyer et al., 2016)
0 > 1.55
1 0/2
2 between 1.56 and 1.60
3 2/4
4 >1.61

218. medial tibial facet height ratio (MTH1/MTH2 in Boyer et al., 2016)
0 <-0.151
1 between -0.15 and 0
2 between 0 and 0.15
3 between 0.151 and 0.30
4 >0.301

219. astragalar neck length (astragalar neck length/mid-trochlear width; contains information like that of RA5-273)
0 <100
1 0/2
2 >100, <120
3 2/4
4 >120

220. astragalar neck angle (Gebo et al., 2001, character 3)
0 <20 degrees
1 0/2
2 20-30 degrees
3 2/4
4 >30 degrees

221. relative astragalar head width
0 head width/head height x 100 <115
1 0/2
2 head width/head height x 100 >115, <130
3 2/4
4 head width/head height x 100 >130

222. plantarflexion of astragalar head*
0 absent
1 present

223. posterior astragalar shelf (D20; contains information like RA4-272)
0 absent
1 0/2
2 present, small
3 2/4

4 present, large

224. fibular facet angle
0 79 degrees or less
1 80-89 degrees
2 90-99 degrees
3 100-109 degrees
4 110 degrees or higher

225. size of lateral tubercle buttressing flexor fibularis
0 poorly developed
1 0/2
2 small
3 2/4
4 large

226. position of flexor hallucis longus groove (Yapuncich et al., 2017)
0 > -0.2
1 between -0.21 and -0.4
2 between -0.41 and -0.6
3 > -0.61

227. depth of flexor hallucis longus groove (Yapuncich et al., 2017)
0 < 1.0
1 between 1.01 and 1.25
2 between 1.26 and 1.5
3 > 1.5

228. calcaneal width/length
0 <35
1 >35, <40
2 >40, <45
3 >45,<50
4 >50

229. anterior calcaneal elongation (length of calcaneus distal to ectal facet/total calcaneal length x 100) (modified from RC1-275, scoring from D22 addendum)
0 >30,<40
1 >41,<50
2 >51,<60
3 >61,<70
4 >71

230. proximal calcaneal elongation (length from proximal border of ectal facet to proximal end of calcaneus/calcaneus length) from Gebo et al. 2001
0 .10-.15
1 .16-.20
2 .21-.25
3 .26-.30
4 >.31

231. relative length of ectal facet (ectal facet width/ectal facet length)
0 <50
1 0/2
2 >50, <60
3 2/4

4 >60

232. ectal facet radius of curvature
0 <120 degrees
1 >=120 degrees

233. ectal facet flange development
0 absent
1 intermediate
2 present

234. bony distinction between plantar edge of ectal facet and body
0 no well-defined border
1 well-defined border present

235. size of calcaneal peroneal tubercle (100*widest point on peroneal tubercle to ectal facet, measured perpendicular to long axis of bone/calcaneal width)
0 massive, extends far laterally (>=21)
1 medium (<21)
2 highly reduced or too small to measure

236. peroneal tubercle shape
0 discrete tubercle with clearly defined borders
1 tubercle gradually slopes into body (substantially longer proximodistally relative to mediolateral width)

237. number of "peaks" on calcaneal peroneal tubercle
0 single tubercle present
1 two "peaks" present

238. position of peroneal process relative to sustentaculum
0 below
1 lower half
2 upper half

239. position of the peroneal tubercle relative to ectal facet (RC2-276)
0 maximum width of peroneal tubercle is placed distal to distal terminus of ectal facet
1 maximum width of peroneal tubercle is placed approximately at the distal terminus of the ectal facet
2 maximum width of peroneal tubercle is placed proximal to the distal terminus of the ectal facet, closer to the distal terminus of the ectal facet than the proximal
3 maximum width of peroneal tubercle is placed proximal to distal terminus of the ectal facet, closer to the proximal terminus of the ectal facet than distal

240. development of distal plantar tubercle on calcaneus
0 small, poorly developed
1 0/2
2 well-developed

241. level of distal plantar tubercle (tubercle to cuboid facet/calcaneal length)
0 distal, near cuboid facet (0-0.19)
1 more proximal (>0.20)

242. orientation of long axis of calcaneocuboid joint
0 dorsoventral
1 oblique
2 mediolateral

243. calcaneocuboid joint shape
    0 oval
    1 fan-shaped, notch is articular
    2 fan-shaped, notch is non-articular
    3 oval but longest in dorsoplantar direction

244. concavity of cuboid facet
    0 flat (>=160
    1 shallow (>=150, <160
    2 deep (<150)

245. dorsal extension of bone supporting cuboid facet
    0 absent
    1 present

246. posterior (plantar) calcaneal bowing
    0 absent
    1 present, moderately developed
    2 present, extreme bowing

247. mediolateral bowing of calcaneal tuber
    0 absent
    1 present, moderate bowing
    2 present, extreme bowing

248. calcaneal sustentacular facet configuration
    0 single continuous sustentacular facet present
    1 0/2
    2 separate anterior and posterior sustentacular facets present

249. form of medial margin between sustentaculum and anterior calcaneal facet (viewed dorsally)
    0 waisted into body
    1 relatively linear medial margin

250. facet anterior to the anterior extension of the sustentacular facet (navicular or otherwise)
    0 prominent
    1 very small or absent

251. angle between sustentaculum and ectal facet
    0 <140
    1 140-160
    2 >160-<180
    3 >180

252. shape of medial edge of sustentaculum in plantar view
    0 triangular or beak-like
    1 rounded

253. morphology of groove for flexor fibularis under sustentaculum
    0 no noticeable groove present
    1 deep medially with distinct groove or wall

254. position of distal terminus of ectal facet relative to anterior calcaneal segment
    0 dorsally positioned on body
    1 intermediate condition
    2 sunk into body plantarly

255. size of tuber calcanei
    0 relatively small (sliver, oval, no clear insertion for tendo calcanei)
    1 medium (large in one direction, insertion for tendo calcanei present but not robust)
    2 relatively large

256. shape of tuber calcanei
    0 oval
    1 square/rectangular
    2 uneven walls/trapezoidal
    3 proximally projecting medial wall

257. dorsal heel process
    0 tucked under ectal facet in lateral view
    1 level with ectal facet in lateral view
    2 projects dorsal to ectal facet

258. navicular length relative to width (contains information like RN1-278)
    0 (<100)
    1 (100-150)
    2 (>150% longer than wide)

259. 'three-pronged' distal navicular articulation
    0 absent
    1 present

260. morphology of the naviculocuboid articulation (RN3-279)
    0 cuboid facet contacts only the ectocuneiform
    1 cuboid facet contacts the ectocuneiform and the mesocuneiform

261. MTI relative dorsoplantar diaphysis midshaft breadth
    0 wide
    1 narrow

262. MTI proximal articular surface angle, in dorsolateral view with respect to the proximal end of the bone
    0 <=105 degrees
    1 =>105 degrees

263. MTI proximal articular surface shape in dorsolateral view
    0 slight concavity
    1 deep concavity, saddle-shaped

264. MTI proximal articular surface torsion
    0 <0
    1 0-19
    2 20-39
    3 40-59
    4 60-79

265. MTI peroneal process proximodistal length in dorsal view with respect to the distal end of the bone
    0 flat
    1 short
    2 intermediate
    3 long

266. MTI peroneal process dorsoplantar thickness in dorsolateral view with respect to the proximal end of the bone

0 flat
1 shallow
2 deep

267. MTI medial side of proximal articular surface with respect to the proximal end of the bone
0 rounded or weak keel
1 strong keel

268. MTI dorsal notch located proximal to the distal articular surface in dorsal or lateral view
0 deep
1 intermediate
2 shallow

269. MTI head distal projection in dorsal view
0 short
1 long

270. MTI head shape in distal view
0 ovoid
1 rectangular

271. MTI distal epicondyle asymmetry in distal and dorsal views
0 medial epicondyle larger
1 epicondyles symmetrical
2 lateral epicondyle larger

272. MTI dorsoplantar axis of head orientation relative to long axis of diaphysis in plantar view, with respect to the distal end of the bone
0 medial
1 intermediate
2 lateral

273. MTI proximal articular surface notch location
0 absent
1 medioplantar
2 lateroplantar

274. MTI distal articular surface midline keel
0 single
1 bifurcate

275. MTI medial proximally projecting tubercle size
0 smaller or equal in size to peroneal process
1 larger than peroneal process

276. MTI dorsal extension of proximal articular surface
0 present
1 absent

277. lateral process of entocuneiform (RE2-281)
0 small
1 hypertrophied

278. foot axis (R01-282)
0 mesaxonic
1 paraxonic

2 ectaxonic

279. metatarsus length (MT3 length/proximal phalanx of digit III) (contains information like that of RO4-285)
0 <1.0
1 >1.01, <1.5
2 >1.51, <2.0
3 >2.01

280. length of MT1 relative to MT3 (RMT2)
0 less than 70% the length of MT3
1 71-80%
2 81-90%
3 91-100%
4 >100%

281. tibial process for peroneus longus
0 absent or poorly developed
1 present and distinct

282. retroflexion of proximal tibial articulation
0 absent, articular surface of tibial condyles perpendicular to long axis of tibial shaft
1 moderate retroflexion present
2 proximal articular surface is strongly retroflexed

283. fusion of tibia and fibula (RT1-263)
0 absent, small articulation
1 no fusion, moderate articulation
2 no fusion but extensive articulation
3 fusion

284. shape of distal surface of tibia (RT3-264)
0 square
1 intermediate
2 triangular

285. rotation of the medial malleolus (RT4-265)
0 none
1 0/2
2 slight
3 2/4
4 strong

286. shape of the medial malleolar articular surface (RT5-266)
0 flat
1 0/2
2 anteriorly convex, posteriorly flat
3 2/4
4 all convex

287. shape of distal tibial shaft (RT6-267)
0 no compression
1 anteroposteriorly compressed

288. position of tibialis posterior groove (RT7-268)
0 on medial side of malleolus
1 0/2

2 on posterior side of malleolus

289. fovea capitis (Y75)
0 small and insignificant
1 0/2
2 present and well-developed

290. femoral head shape (RF6-256)
0 spherical
1 semicylindrical
2 cylindrical

291. morphology of femoral head articular surface
0 restricted to femoral head
1 extends onto back of femoral neck

292. femoral neck length
0 <=75
1 0/2
2 >75, <120
3 2/4
4 =>120

293. angle of femoral neck (RF2-252)
0 <60
1 0/2
2 60-70
3 2/4
4 >70

294. anteroposterior angulation of proximal femur (contains information like RF8-258)
0 bent anteriorly
1 not bent anteriorly

295. triangular depression on medial side of proximal femur (Dagosto and Schmid, 1996)
0 absent
1 present

296. greater trochanter shape (Y72)
0 hook-shaped
1 0/2
2 rounded

297. crista paratrochanterica
0 absent
1 0/2
2 present

298. relative length of trochanteric fossa (RF9-259)
0 >125 (long)
1 0/2
2 110-125 (moderate)
3 2/4
4 <110 (very short)

299. intertrochanteric crest (RF10-260)

0 absent
1 present

300. angle of lesser trochanter (RF3-253)
0 medial (0-30 degrees)
1 0/2
2 posterior (>30 degrees)

301. gluteal tuberosity (Y74, contains information like RF4-254)
0 present, large
1 0/2
2 present, small and reduced to a thin crest
3 2/4
4 absent

302. position of gluteal tuberosity
0 proximal to lesser trochanter
1 0/2
2 at level of lesser trochanter
3 2/4
4 distal to lesser trochanter

303. patellar margin asymmetry (raised and sharply defined lateral patellar margin) (contains information like RF12-262)
0 absent
1 present

304. knee index (contains information like RF5-255)
0 100 or higher
1 90-99
2 80-89
3 70-79

305. shape of manubrium (Y83)
0 sternal head elongate
1 shortened and triangular

306. transpedicular foramina (Y85)
0 absent
1 present

307. number of thoraco-lumbar vertebrae
0 17 or fewer
1 18
2 19
3 20
4 21 or more

308. anterior inferior iliac spine (Y70)
0 tuberosity not expressed
1 small tubercle present along margo acetabuli
2 well-developed tuberosity present

309. ilium shape (Y67)
0 narrow
1 0/2

2 markedly flared

310. ischial flaring (Y69)
0 absent
1 present

311. ischium/ilium index (Anemone & Covert, 2000)
0 60-70
1 50-59
2 40-49
3 30-39 or <

312. pubic bone flaring (Y68)
0 absent
1 present

313. tail length (Y86)
0 long tail
1 reduced
2 absent

314. shape of glenoid fossa (RCr40-222)
0 wide and biconcave
1 anteroposteriorly oriented trough

315. postglenoid process (Y25)
0 present
1 absent

316. postglenoid process-auditory bulla relationship (Y26)
0 not confluent
1 0/2
2 partially fused
3 2/4
4 postglenoid process fused to lateral aspect of bulla

317. postglenoid foramen
0 present
1 absent

318. position of postglenoid foramen relative to postglenoid process
0 posterior to postglenoid process
1 0/2
2 medial to postglenoid process

319. entoglenoid process (RCr41-223)
0 indistinct or absent
1 0/2
2 robust

320. position of pyramidal processes (RCr35-217)
0 medially placed
1 0/2
2 laterally placed

321. position of the anteriormost point on the palatine/maxillary suture of the palate (Y19)

0 medial to space between M1 and M2 or more posterior
1 0/2
2 medial to M1
3 2/4
4 medial to the space between P4 and M1

322. posterior palatine torus (RCr34-216)
0 present
1 0/2
2 absent

323. maxillary-palatine notch
0 absent
1 present

324. position of the posteromedial edge of the palate in relation to M3 (Y24)
0 anterior to M3
1 0/2
2 medial to M3
3 2/4
4 posterior to M3

325. shape of medial pterygoid plate (RCr36-218, contains information like Y22)
0 long medial pterygoid plate extending one-third to one half of the distance to the anterior surface of the bulla
1 0/2
2 short but distinct from lateral pterygoid plate for its entire dorsoventral extent
3 2/4
4 medial pterygoid plate entirely absent, or reduced to a low rugosity

326. lateral pterygoid-bullar overlap (RCr18-200)
0 absent
1 0/2
2 abutting
3 2/4
4 laminar

327. extent of contact between the lateral pterygoid plate and the bullar wall (RCr19-201)
0 slight
1 very extensive

328. position of foramen ovale (Y28)
0 lateral side of lateral pterygoid
1 0/2, or posterior to terminus of lateral pterygoid
2 medial to lateral pterygoid

329. central stem of basicranium (D47)
0 narrow
1 broad

330. choanal shape (D50)
0 broad
1 peaked

331. basioccipital flange (contains information like RCr20-202)
0 absent or minimal

1 present, extensive

332. encroachment of auditory bulla on pterygoid fossa (RCr17-199)
0 absent
1 present and formed by the AAC
2 present and formed by the tympanic cavity
3 1/2

333. shape of external auditory meatus (Fleagle & Kay, 1983)
0 tubular
1 0/2
2 not tubular

334. suprameatal foramen (RCr21-203)
0 absent
1 present, small, and in the posterior root of the zygomatic arch
2 present, large, and above the external auditory meatus
3 0/2

335. ascending pharyngeal artery (Y52)
0 if present is poorly developed
1 present and enlarged

336. vascular plexus associated with ascending pharyngeal artery (Y53)
0 absent
1 present

337. position of chorda tympani relative to m. tensor tympani (Maier, 2008)
0 ventral to m. tensor tympani
1 dorsal to m. tensor tympani
2 m. tensor tympani absent

338. relative sizes of stapedial canal (SC) and promontory canal (PC)
0 LN(PC diameter/SC diameter) <0
1 LN(PC diameter/SC diameter) >0.01, <1.0
2 LN(PC diameter/SC diameter) >1.01, <2.0
3 LN(PC diameter/SC diameter) >2.01, <10
4 LN(PC diameter/SC diameter) >10

339. mediolateral position of posterior carotid foramen (RCr6-188)
0 medial
1 midline of bulla
2 lateral

340. rostrocaudal position of posterior carotid foramen with respect to fenestra cochleae (contains information like RCr8-190)
0 posterior to fenestra cochleae
1 ventral to fenestra cochleae
2 anterior to fenestra cochleae

341. ventrodorsal position of posterior carotid foramen relative to fenestra cochleae (RCr7-189)
0 dorsal
1 ventral

342. position of pathway for internal carotid artery or nerve relative to fenestra cochleae (RCr9-191)
0 runs across ventral lip of fenestra cochleae, shielding it from ventral view

1 does not shield fenestra cochleae

343. morphology of path for promontory artery and/or internal carotid nerve (RCr12-194)
0 promontorial canal present and fully enclosed in its course through the middle ear
1 promontorial canal present but short and continuous anteriorly with trough/groove
2 promontorial canal absent, trough or groove present across entire promontorium

344. perbullar pathway for internal carotid artery (RCr4-186)
0 absent
1 present

345. transverse septum defining caudal wall of anterior accessory cavity (contains information like RCr1-183)
0 absent
1 present, forms lateral wall of AAC pneumatized from tympanic cavity
2 present, forms lateral wall of AAC pneumatized from epitympanic recess

346. trabeculated anterior accessory cavity
0 absent
1 present

347. number of cochlear turns
0 <1.9
1 2.0-2.49
2 2.5-2.99
3 >3

348. composition of bulla
0 petrosal
1 entotympanic
2 ectotympanic + entotympanic

349. epitympanic crest (RCr48-230)
0 absent
1 present

350. morphology of annular bridge (RCr16-198)
0 linea semicircularis or partial anular bridge formed on an entotympanic bulla
1 linea semicircularis formed on a petrosal bulla
2 complete annular bridge

351. position of ventral edge of tympanic bone (RCr14-196)
0 intrabullar
1 extrabullar

352. parotic fissure (RCr22-204)
0 patent
1 closed

353. pneumatization of mastoid (from epitympanic recess) (RCr3-185)
0 absent
1 present, moderately inflated
2 present, greatly inflated

354. paroccipital processes (Y23)
0 absent
1 0/2

2 present, small
3 2/4
4 present, well-developed

355. emissary foramina on lateral edge of parietals (Y33)
0 present
1 0/2
2 absent

356. pattern of sagittal cresting
0 temporal lines converge on frontal
1 temporal lines converge on the parietals, forming short sagittal crest
2 temporal lines are not confluent, no sagittal crest present
3 sagittal crest is extensive, but temporal lines converge on parietals that extend far rostrally between frontal bones
4 0/2

357. metopic sutural fusion in adults (RCr31-213)
0 absent
1 0/2
2 present

358. frontal-maxillary contact
0 present
1 absent due to intervening lacrimal
2 absent due to intervening premaxilla

359. degree of orbital convergence (contains information like RCr32-214)
0 >89
1 90-104
2 105-119
3 120-134
4 >135

360. expansion of ethmoturbinals (YIP89)
0 no anteromedial expansion
1 anterior and medial expansion

361. transverse lamina/ethmoturbinal recess
0 extensive transverse lamina and ethmoturbinal recess present
1 extensive transverse lamina and ethmoturbinal recess absent

362. pronounced interorbital constriction (RCr27-209)
0 absent
1 present below CN I

363. orbit size (contains information like RCr23-205)
0 small
1 large
2 hypertrophied

364. foramen rotundum (Y12,RCr29-211)
0 confluent with the superior orbital fissure
1 present

365. lacrimal foramen morphology (Y11)

0 surrounded by lacrimal bone
1 foramen lies on lacrimal/maxillary suture
2 foramen lies on the lacrimal/maxillary suture but is primarily surrounded by maxilla

366. position of lacrimal foramen (RCr30-212)
0 on rim, or outside of, orbit
1 inside the orbit

367. zygomatic-lacrimal contact (Y6, RCr26-208)
0 present
1 0/2
2 absent

368. lacrimal-palatine contact (RCr28-210)
0 contact present
1 no contact, separated by a large frontal-maxillary contact (and in some taxa, a small os planum of the ethmoid)
2 no contact, separated by a large os planum

369. ethmomaxillary fissure (Y13, modified following Cartmill & Kay, 1978)
0 no complete ethmomaxillary fissure present
1 0/2
2 complete ethmomaxillary fissure present

370. anterior and posterior hiatus of ethmomaxillary fissure (Cartmill & Kay, 1978)
0 anterior hiatus variably present
1 0/2
2 posterior hiatus at least variably present
3 2/4
4 no hiatuses present

371. frontal-palatine contact (Cartmill & Kay, 1978)
0 present
1 0/2
2 absent

372. position of sphenopalatine foramen
0 within palatine
1 0/2
2 on ethmopalatine suture

373. position of infraorbital foramina (contains information like Y3)
0 above P2
1 above P3
2 above P4
3 above M1

374. morphology of zygomatic portion of postorbital bar (Y18, contains information like RCr24-25)
0 no postorbital bar
1 slender and delicate
2 1/3
3 wide and robust
4 contributes to postorbital septum

375. morphology of frontal segment of the postorbital bar (Y17, contains information like RCr24-25)
0 partial frontal process but no postorbital bar

1 slender and delicate
2 1/3
3 wide and robust
4 incorporated into postorbital septum

376. zygomatic arch depth
0 slender
1 0/2
2 dorsoventrally deep

377. zygomaticofacial foramen (Y16)
0 absent
1 0/2
2 present, small
3 2/4
4 present, large

378. snout length (RCr37-219)
0 long snout
1 short snout

379. maxillary depth (RCr38-220)
0 deep
1 shallow

380. facial profile (Masters & Brothers, 2002)*
0 straight
1 0/2
2 dished

381. position of anterior palatine foramina relative to incisors (Masters & Brothers, 2002)*
0 foramina sit behind first incisors
1 0/2
2 foramina intrude between first incisors

382. rostral projection of premaxilla (Masters & Brothers, 2002)*
0 no extension of premaxilla beyond anterior incisors
1 mild premaxillary projection
2 considerable projection, forming a tubular anterior rostrum

383. ascending wing of premaxilla (RCr49-231)
0 narrow
1 broad

384. retia mirabilia of the proximal limb vessels (YIP87)
0 absent
1 present

385. nasal morphology (Y62, contains information like R289)
0 lateral cleft between medial and lateral nasal processes
1 lateral cleft fused and rhinarium covered with dry hairy skin

386. retinal fovea (Y60, R288)
0 absent
1 present

387. tapetum lucidum (Y59, R290)
0 present
1 absent

388. presence/absence of choriovitelline placenta (Y58, KPL4)
0 choriovitelline placenta
1 no choriovitelline placenta

389. epitheliochorial vs. hemochorial placentation (KPL1)
0 epitheliochorial
1 diffuse, hemochorial

390. blastocyst attachment (KPL2)
0 invasive
1 non-invasive

391. amniotic cavity (KPL3)
0 primordial cavity absent
1 primordial cavity present

392. embryonic body stalk (KPL5)
0 absent
1 present

393. allantois development (KPL6)
0 large, vesicular
1 rudimentary

394. ascorbic acid synthesis
0 absent
1 present

395. Maxillary sinus
0 present
1 absent

396. ischial callosity
0 absent
1 present

397. medial postentocristid m1-m2
0 absent or weak and rounded
1 strong, ruins towards the hypoconid

398. medial posterior transverse crista
0 absent, or weak and rounded
1 strong, directed towards hypocone

399. Alisphenoid contact and “postorbital closure”
0 alisphenoid short, does not send process into the periorbital septum
1 alisphenoid invades the periorbital septum, reaches and sutures with the zygomatic walling off the orbit from the temporal fossa

404. Orientation of the nasolacrimal duct
0 obliquely oriented
1 more vertical

Added notes on new and revised characters of Seiffert matrix (83)

Seiffert character 363.

Comment: This character covers orbital hypertrophy but lists *Shoshonius* as hypertrophied. However, in a log-log plot of orbit area vs. prosthion-Inion length, Shoshonius falls on the line with strepsirrhines NOT with hypertrophied tarsiids and *Aotus*.

Seiffert Character 374. Morphology of zygomatic portion of postorbital bar (Y18 (105)), contains information like RCr24-25 (116). Revised as follows:

0 no postorbital bar
1 slender and delicate
2 1/3
3 wide and robust
4 contributes to postorbital septum

REVISED Seiffert Character 374. Presence or absence of a frontal- zygomatic contact producing a lateral orbital bar (Y18 (105)), contains information like RCr24-25 (116)). Revised as follows:

0 frontal does not contact zygomatic; no postorbital bar
1 frontal contacts zygomatic producing a postorbital bar

Seiffert Character 375. morphology of frontal segment of the postorbital bar (Y17(105)), contains information like RCr24-25 (116)) . Revised as follows:

0 partial frontal process but no postorbital bar
1 slender and delicate
2 1/3
3 wide and robust
4 incorporated into postorbital septum

REVISED Seiffert Character 375. Contributions of zygomatic and frontal to postorbital closure (RCr 25 (116). Revised as follows:

0 zygomatic predominant
1 frontal predominant

Comment: Scored as ? when there is no P-O closure.

New Character 395:maxillary sinus.

0 present
1 absent

Comment: Separates cercopithecoids, which lack a maxillary sinus, from all other euprimates, see Koppe and Nagai (117)

New Character 396: Ischial callosities.

0 absent
1 present

Comment: Separates cercopithecoids from all other euprimates, see McCrossin and Benefit (118)

New Character 397: Medial postentocristid m1-m2.

0 absent
1 present

Comment: Separates cercopithecoids which have a medial postentocristid, from all other euprimates). See Kay (119).

New Character 398. Medial posterior transverse crista:

0 absent;
1 present.

Comment: Separates cercopithecoids which have a posterior transverse crista, from all other euprimates). See Kay (119).

New Character 399. Alisphenoid contact and 'postorbital; closure':
0 alisphenoid short, does not send process into the periorbital septum;
1 alisphenoid invades the periorbital septum, reaches and sutures with the zygomatic walling off the orbit from the temporal fossa.
Comment: All Plesiadapiformes, crown strepsirrhines and all Eocene primates have state 0; tarsiers and anthropoids have state 1. All euprimates have a bony postorbital bar running from the braincase to the zygomatic arch, [postorbital bar formed by contact between frontal and zygomatic] but anthropoids and tarsiers are distinguished from all other primates (and mammals) by the expansion of this bar into a postorbital septum-- a thin wall of bone separating the orbit from the temporal fossa. The postorbital septum is composed of the same bones in both tarsiers and anthropoids and although the relative contributions of these bones vary between tarsiers and anthropoids (120), the alisphenoid always contacts the zygomatic, a feature unique to tarsiers and anthropoids among all vertebrates (121). The presence of the postorbital septum acts to reduce the bulging of the eye when the muscles of mastication contract, so it may be viewed as an adaptation for enhanced visual acuity (122).

New Character 404. Orientation of the nasolacrimal duct.
0 obliquely oriented;
1 more vertical.
Comment: A haplorhine character said to be independent of the size of the rostrum (104, 123-124).

Characters 401-403 identified in the course of this work but not used in the phylogenetic analysis

New Character 400. Relative size of the optic foramen:
0 very small;
1 small;
2 large.
Comment: The foramen transmits the optic nerve and ophthalmic artery. Its size reflects the size of the nerve, which in turn reflects the number of nerve fibers and the extent of retinal summation. Operationally we compare Ln optic foramen size with body size, as expressed by Ln skull length (prosthion-Inion) in a log-log plot. We don't use Optic Foramen Index because that is a measure of the size of the optic foramen compared with orbit size. Plesiadapiformes have a very small foramen (state 0); it is larger (state 1) in strepsirrhines, omomyoids and adapoids reflecting a transition to improved vision; it is very large in tarsiers and anthropoids (state 2) indicating heightened visual acuity (reduced retinal summation, especially in the central retinal area. We note in passing that the Middle Eocene *Shoshonius* has a foramen size like that of other omomyoids.

New Character 401. Size of the neocortex.
0 small;
1 large;
2 very large.
Comment: As measured on Ln neocortex surface area vs. Ln brain volume on endocasts see this paper. Also as indicated by endocast landmarks—exposure of midbrain, as noted by Long, Bloch & Silcox (35) and underlying anatomical data from Stephan et al. General expansion of the neocortex in Euprimates compared with Plesiadapiformes and further expansion in crown haplorhines, perhaps related to increased importance of the visual system, Plesiadapiformes (0) vs. Eocene euprimates and extant strepsirrhines (1) versus tarsiers and anthropoids (2)

New Character 402. Proportion of the neocortex taken up by the combined occipital, parietal, and temporal lobe (OPT) surface areas compared with endocast volume.
0 OPT lobes not distinguishable from frontal lobe;
1 sylvian fissure present with small OPT component;
2 sylvian fissure present with large OPT component.
Comment: The proportions of OPT neocortex may indicate the extent of the neocortex devoted to visual processing. Plesiadapiformes (0) vs Eocene euprimates and extant strepsirrhines (1) versus tarsiers and anthropoids (2). Data presented in this paper. Presence of a discrete temporal lobe of the neocortex (from endocasts) was noted by Le Gros Clark (43).

New Character 403. Volume of the olfactory bulb compared with volume of brain (from endocasts):
0 Olfactory bulb very large;
1 Olfactory bulb large;
2) Olfactory bulb small.
Comment: Olfactory bulb is very large in Plesiadapiformes (e.g., *Ignacius*, *Microsyops*); it is large in Eocene primates and crown strepsirrhines; 2) Olfactory bulb small in crown haplorhines. Character 1 takes in a lot of real estate, see Heritage (125). There is some wiggle room for an added character since *Simonsius* and *Rooneyia* may have a larger bulb than crown anthropoids, but on the edge of the extant strepsirrhine range, see Kirk et al. (104).